



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to S.P. SAMY AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 91.54 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (11.457178, 78.529823)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route starts at CVQF+23W in Sangagiri and proceeds to FG5J+432 in Salem via the Tiruchengode - Salem Main Road and the village of Thokkavadi. This journey spans approximately 91.54 kilometers. The

route primarily consists of state highways and rural roads, incorporating several small towns and agricultural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with a significant amount of rainfall from the northeast monsoon (October to December). During these months, heavy rainfall could lead to waterlogging and reduced visibility. Conversely, the summer months (March to June) often see extreme heat, potentially causing road surface damage and increasing vehicular overheating risks.

3. Analysis of Traffic Patterns

Traffic tends to be heavier during the morning (8:00-10:00 AM) and evening (5:00-7:00 PM) rush hours, especially around urban areas like Salem. Notable congestion points include the Tiruchengode market areas and intersections near Salem city limits. Heavy vehicle operators should anticipate slower travel during these periods.

4. Assessment of Road Quality and Infrastructure

The road quality varies, with better-maintained stretches on the state highways and less-developed sections through rural areas. Narrow lanes, potholes, and lack of proper signage are common concerns on secondary roads. Frequent road repairs can also cause temporary bottlenecks.

5. Suggestions for Alternative Routes for Emergencies

In case of congestion or roadblocks, an alternative route could involve taking State Highway 86 (SH-86) via Namakkal before connecting back to the main road toward Salem. This detour might be longer but can be useful if the primary route is compromised.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict regulations on the transport of hazardous materials, including mandated escort vehicles for certain classes of dangerous goods. Compliance with these laws is mandatory, and trucks should be equipped with proper placarding and documentation.

7. Overview of Historical Incidents

The route generally has a low incidence of major accidents involving heavy vehicles or hazardous materials. However, minor incidents related to vehicular collisions due to inadequate road conditions have been reported, often influenced by adverse weather.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural zones and small settlements. Particular care should be taken to avoid contaminating water bodies and farmland in case of spills. No significant wildlife habitats or

ecological reserves are immediately affected by the route.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable along this route with some possible dead zones in rural areas. Coverage may vary between mobile network providers, so having multiple communication options is advisable.

10. Estimated Emergency Response Times

Emergency response times can vary, with faster responses possible within urban areas like Salem (approximately 30-45 minutes). In rural locales, especially during bad weather or peak traffic periods, response times could exceed one hour.

11. An Overall Summary of Risk Assessment

Overall, the route is moderately challenging, presenting typical hazards associated with regional roads in Tamil Nadu. Weather conditions and traffic can significantly impact travel time and safety, particularly for vehicles carrying hazardous materials. Proper planning, adherence to regulations, and situational awareness are crucial for mitigating risks.

The route analysis provides essential insights for training truck drivers to navigate this path efficiently while ensuring safety and adherence to regional guidelines.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38346, 77.89975	30 KM/Hr
5	Turn	Medium	11.38302, 77.89995	30 KM/Hr
6	Turn	High	11.38188, 77.90257	15 KM/Hr
7	Turn	High	11.37830, 77.90160	15 KM/Hr
8	Turn	Medium	11.36754, 77.93919	30 KM/Hr
9	Turn	Medium	11.41361, 78.08073	30 KM/Hr
10	Turn	High	11.41453, 78.08062	15 KM/Hr
11	Turn	Medium	11.41476, 78.08094	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.42956, 78.11590	15 KM/Hr
13	Turn	Medium	11.42185, 78.13013	30 KM/Hr
14	Turn	Medium	11.42450, 78.15278	30 KM/Hr
15	Turn	Medium	11.43432, 78.20586	30 KM/Hr
16	Turn	High	11.43542, 78.20754	15 KM/Hr
17	Turn	Medium	11.45075, 78.20391	30 KM/Hr
18	Turn	High	11.45099, 78.20397	15 KM/Hr
19	Turn	High	11.44946, 78.22270	15 KM/Hr
20	Turn	High	11.44793, 78.22320	15 KM/Hr
21	Turn	Medium	11.44753, 78.22443	30 KM/Hr
22	Turn	Medium	11.45361, 78.23597	30 KM/Hr
23	Turn	Medium	11.49628, 78.32342	30 KM/Hr
24	Turn	Medium	11.50868, 78.33972	30 KM/Hr
25	Turn	High	11.51180, 78.34422	15 KM/Hr
26	Turn	Medium	11.51040, 78.34875	30 KM/Hr
27	Turn	Medium	11.50969, 78.34967	30 KM/Hr
28	Turn	Medium	11.50905, 78.35285	30 KM/Hr
29	Turn	Medium	11.48662, 78.36896	30 KM/Hr
30	Turn	Medium	11.47977, 78.38044	30 KM/Hr
31	Turn	Medium	11.47754, 78.38114	30 KM/Hr
32	Turn	Medium	11.46956, 78.39365	30 KM/Hr
33	Turn	Medium	11.46783, 78.39421	30 KM/Hr
34	Turn	High	11.46455, 78.39289	15 KM/Hr
35	Turn	High	11.45614, 78.40096	15 KM/Hr
36	Turn	Medium	11.45560, 78.40272	30 KM/Hr
37	Turn	Medium	11.45658, 78.40442	30 KM/Hr
38	Turn	Medium	11.45483, 78.40829	30 KM/Hr
39	Turn	Medium	11.46741, 78.45087	30 KM/Hr
40	Turn	High	11.46729, 78.45447	15 KM/Hr
41	Turn	Medium	11.46808, 78.45535	30 KM/Hr
42	Blind Spot	Blind Spot	11.46798, 78.45844	10 KM/Hr
43	Blind Spot	Blind Spot	11.46857, 78.45844	10 KM/Hr
44	Turn	High	11.46799, 78.45992	15 KM/Hr
45	Turn	Medium	11.46458, 78.47401	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	11.46418, 78.47451	30 KM/Hr
47	Turn	Medium	11.46333, 78.48024	30 KM/Hr
48	Turn	Medium	11.46211, 78.48159	30 KM/Hr
49	Turn	Medium	11.46177, 78.48277	30 KM/Hr
50	Turn	High	11.46097, 78.48362	15 KM/Hr
51	Blind Spot	Blind Spot	11.44226, 78.48592	10 KM/Hr
52	Turn	Medium	11.45575, 78.52476	30 KM/Hr
53	Turn	High	11.45633, 78.52755	15 KM/Hr
54	Turn	High	11.45724, 78.52761	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
10	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
11	hospital	Annai Hospital	11.4437321, 78.4858654	30 km/h	Medium
12	clinic	Shreetejo Ayurveda Clinic	11.444915, 78.485868	30 km/h	Medium
13	hospital	Vinola Hospital	11.4439899, 78.4857585	30 km/h	Medium
14	hospital	The Chirist Hospital	11.4443163, 78.4862134	30 km/h	Medium
15	clinic	Sabeetha Surgical Clinic	11.4421254, 78.4867012	30 km/h	Medium
16	hospital	Dr. Ramu Life Care Hospital	11.4426079, 78.4879762	30 km/h	Medium
17	hospital	Thai Hospital	11.4424776, 78.4894516	30 km/h	Medium
18	clinic	Jayaraj Memorial Clinic	11.4415635, 78.4890421	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38302, 77.89995



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38188, 77.90257



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37830, 77.90160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36754, 77.93919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41361, 78.08073



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41476, 78.08094



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42956, 78.11590



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42450, 78.15278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45075, 78.20391



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45099, 78.20397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44946, 78.22270



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44793, 78.22320



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44753, 78.22443



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45361, 78.23597



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49628, 78.32342



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50868, 78.33972



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.51180, 78.34422



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51040, 78.34875



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50969, 78.34967



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50905, 78.35285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47977, 78.38044



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47754, 78.38114



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46956, 78.39365



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46783, 78.39421



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46455, 78.39289



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45614, 78.40096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45560, 78.40272



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45658, 78.40442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45483, 78.40829



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46741, 78.45087



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46729, 78.45447



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46808, 78.45535



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46798, 78.45844



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46857, 78.45844



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46799, 78.45992



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46458, 78.47401



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46418, 78.47451



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46333, 78.48024



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46211, 78.48159



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46177, 78.48277



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46097, 78.48362

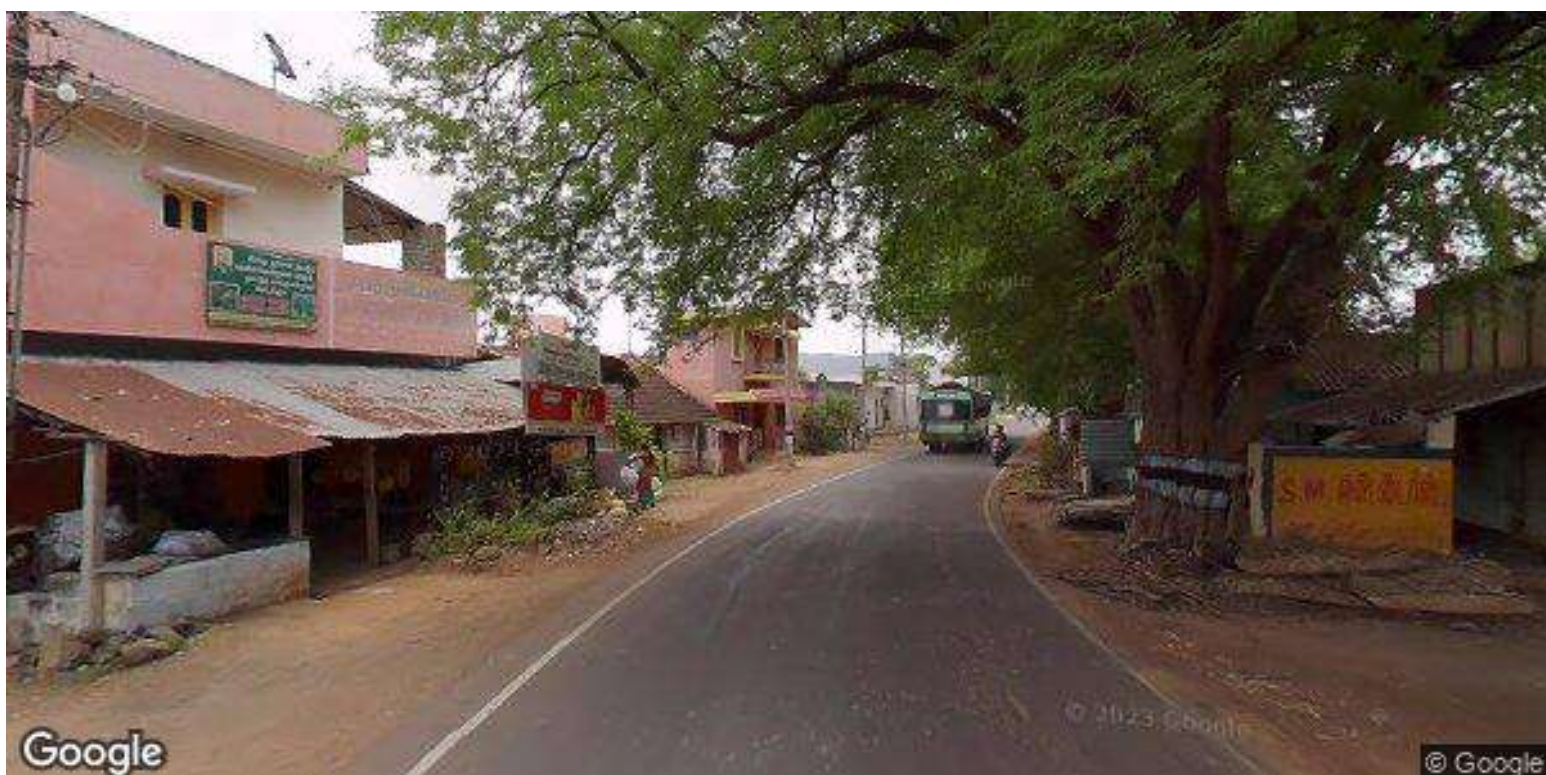


Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44226, 78.48592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45575, 78.52476



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45633, 78.52755



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45724, 78.52761
